25901911 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/MGU-BBA (Honours)/MGU-BCA (Honours) Special Examinations, July 2026 (for 2024 Admission Regular Examinations, October 2025)

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCMAT200 - PERSPECTIVES OF MATHEMATICS

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

{Short Answer Questions}

Answer any **5** questions

Each question carries **2** marks

1. Find the cartesian coordinates of the point whose polar coordinates is (-5,π) [A] / [CO1]

2. Find the parametric equation and parametric interval for the motion of a particle that starts at (a,0) and traces the circle $x^2 + y^2 = a^2$, once counterclockwise. [U] / [CO1]

3. If α, β, γ are the roots of the equation $8x^3 - 4x^2 + 6x - 1 = 0$, obtain the equation whose roots are $2\alpha + 1, 2\beta + 1, 2\gamma + 1$. [A] / [CO4]

4. If q, r, s, are positive, show that the equation $f(x) = x^4 + qx^2 + rx - s = 0$ has one positive, one negative and two imaginary roots. [A] / [CO4]

5. Express the double integral $\int_0^2 \int_0^{\sqrt{4-y^2}} (x^2 + y^2)\, dy\, dx$ in polar coordinates. [K] / [CO5]

6. Write an equivalent integral of $\int_0^3 \int_0^{2x} dydx$ with the order of integration reversed. [A] / [CO5]

7. Define Spherical coordinates of a point in space. [K] / [CO6]

8. Evaluate $\int_{-1}^{1}\int_{0}^{1}\int_{0}^{2}(x+y+z)\,dxdydz.$ [A] / [CO6]

[2x5 = 10]

Part B

{Short Essay Questions}

Answer any **5** questions

Each question carries **6** marks

9. Graph the set of points whose polar coordinates satisfy the conditions $-3 \leq r \leq 2$ and $\theta = \frac{\pi}{4}$. [A] / [CO1]

10. Identify the conic $x^2 - y^2 + 4x - 6y = 6$. Also find the centre, vertices and foci of the conic. [A] / [CO2]

11. Find the equation whose roots are those of $x^4 - 2x^3 + 3x - 5 = 0$ each diminished by 2.. [A] / [CO4]

12. How many real roots are there for the polynomial $x^4 - 1 = 0$. Find all the roots of this equation. [A] / [CO4]

13. Find the area enclosed by the lemniscate $r^2 = 4\cos 2\theta$. [A] / [CO5]

14. Sketch the region bounded by the parabolas $x = y^2 - 1$ and $x = 2y^2 - 2$ Then express the region's area as an iterated double integral and evaluate the integral [A] / [CO5]

15. Evaluate the limits of integration r, θ and z for the solid enclosed by the cylinder $x^2 + y^2 = 4$, bounded above by the paraboloid $z = x^2 + y^2$ and bounded below by the $xy-$plane? [A] / [CO6]

16. Sketch the region of integration and evaluate the integral $\int_{0}^{1}\int_{y}^{1}(x^2 e^{xy})\,dx\,dy.$ [A] / [CO7]

[6x5 = 30]

Part C

{Essay Questions}

Answer any **3** questions

Each question carries **10** marks

17. a.) Given vertices : $(\pm 10, 0)$, e = 0.24 of the ellipse centered at origin of the xy-plane. Find the standard form equation in cartesian coordinates.
b.) Find the eccentricity, foci and directrices of the hyperbola $9x^2 - 16y^2 = 144$. Also graph the hyperbola. [An] / [CO2]

18. Let the roots of $x^3 - 4x^2 + 5x - 2 = 0$ be α, β, γ. [A] / [CO3]

(a) Find α + β + γ, αβ + βγ + γα, αβγ.

(b) Find $\alpha^2 + \beta^2 + \gamma^2$.

(c) Find $1/\alpha + 1/\beta + 1/\gamma$.

19. (i)Form a quartic equation having real coefficients with $1+i$ and $2-\sqrt{3}i$ as roots. [U] / [CO3]

(ii)Solve $x^4 - 7x^3 + 18x^2 - 22x + 12 = 0$ given that $1+i$ is a root.

20. Find $\iint_R (x+y)dxdy$ where R is [A] / [CO5]

1. the region bounded by $x = 0, y = 0, x + y = 1$
2. the region bounded by the first quadrant of the circle $x^2 + y^2 = 9$.

21. Let D be a cube in the first octant bounded by the coordinate planes and $x = 2, y = 2$, and $z = 2$ and Let $f(x, y, z) = x^2 + 9$ is a function over D [A] / [CO6]

(a) Find the volume of D

(b) Find the average value of F over D

22. Find $\iiint |xyz|\, dx\, dy\, dz$ over the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \leq 1$ using the substitution $x = au\,, y = bv\,,\ z = cw.$ [A] / [CO7]

[10x3 = 30]